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Roll No. 3112413042

324551(24)

BE (5th Semester)
Examination, Nov.-Dec., 2015

(New Scheme)

Electrical Machines-II

Time Allowed : 3 hours *Maximum Marks* : 80
Minimum Pass Marks : 28

Note : (*i*) Part *(a)* of each question is compulsory and attempt any **two** parts from (*b*), (*c*) and (*d*).
(*ii*) The figures in the right-hand margin indicate marks.

1. *(a)* Give reasons for short pitching the coil. 2

(b) What is the difference between electrical and mechanical angle? Show that the flux per pole for sinusoidal distributed flux density is

$$\phi = \frac{4}{P} B_p l r$$

where B_p = peak flux density
P = number of poles
l = active coil-side length
r = mean radius of the stator at air-gap 7

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(c) Derive the torque expression for doubly excited system : 7

$$T_e = \frac{1}{2} i_s^2 \frac{dL_s}{d\theta} + \frac{1}{2} i_r^2 \frac{dL_r}{d\theta} + i_s i_r \frac{dM_{sr}}{d\theta}$$

(d) A 3-phase, 16-pole synchronous generator has a star-connected winding with 144 slots and 10 conductors per slot. The flux per pole is 0·04 Wb and the speed is 375 r.p.m. Find the phase and line induced e.m.f.s. The total turns/phase may be assumed to be series connected. 7

2. *(a)* Under what condition does alternator voltage regulation become negative? 2

(b) Explain how the open and short circuit tests are conducted on synchronous machine. 7

(c) Draw the steady-state power angle characteristics of cylindrical rotor synchronous machine and also derive

$$P_e = \frac{E_f V_t}{X_s} \sin \delta$$

7

(d) Derive basic equivalent circuit model of cylindrical rotor synchronous machines and draw the phasor diagrams for motor and generator at lagging, leading and unity power-factor conditions. 7

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3. (a) What is V-curve of synchronous machine? 2

(b) Explain slip test for measuring X_d and X_q of salient pole synchronous machine. 7

(c) Explain the starting methods of synchronous motors. 7

(d) Two 3-phase alternators operate in parallel. The rating of one machine is 50 MW and that of the other is 100 MW. Both alternators are fitted with governors having a drop of 4%. How will the machines share a common load of 100 MW? 7

4. (a) What are the slips at starting and at steady state of an induction motor on no-load? 2

(b) Explain torque-slip characteristics of 3-phase induction motor showing braking, motoring and generating modes. 7

(c) Explain the no-load and block-rotor tests for determination of equivalent circuit parameters of 3-phase induction motor. 7

(d) A 4-pole, 50 Hz, 3-phase induction motor develops a maximum torque of 110 Nm at 1360 r.p.m. The resistance of the star-connected rotor is 0·25 ohm/phase. Calculate the value of resistance that must be inserted in series, with each rotor phase to produce a starting torque equal to half the maximum torque. 7

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5. (a) Why does induction motor draw heavy current during starting? 2

(b) Discuss in detail the starting methods of 3-phase induction motors. 7

(c) Explain : 7

(i) Cogging

(ii) Crawling

(d) A 3-phase squirrel-cage induction motor has maximum torque equal to twice the full-load torque. Determine the ratio of motor starting torque of its full-load torque, if it is started by (i) direct-on-line and (ii) auto-transformer starting with 80% tapping.

The per phase rotor resistance and per phase standstill reactance ref. to stator side are 0·2 Ω and 2 Ω respectively. Neglect stator impedance. 7

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